

Chapter 1

1.1 Basic Principle of Electricity

1.1.1 Electricity

Electricity is the flow of electrical energy. Electricity is the secondary form of energy which means that other sources of energy i.e., kinetic energy, thermal energy and solar energy etc. are converted into electrical energy.

Do you know?

- In dams, kinetic energy of water is converted into electrical energy.

Point to Ponder

- Which type of energy is converted into electrical energy in coal-based power plants?

How does electricity get to your home?

Electricity is generated at dams and thermal power plants located in remote areas. The electricity that power plants generate is delivered to consumers over transmission and distribution power lines. This complex system sometimes called the “grid” includes substations, transformers, and power lines that connect electricity producers and consumers.

1.1.2 Electrical Quantities and Units

Electricity is represented by electricity quantities. There are three main electrical quantities:

- I. Voltage
- II. Current
- III. Resistance

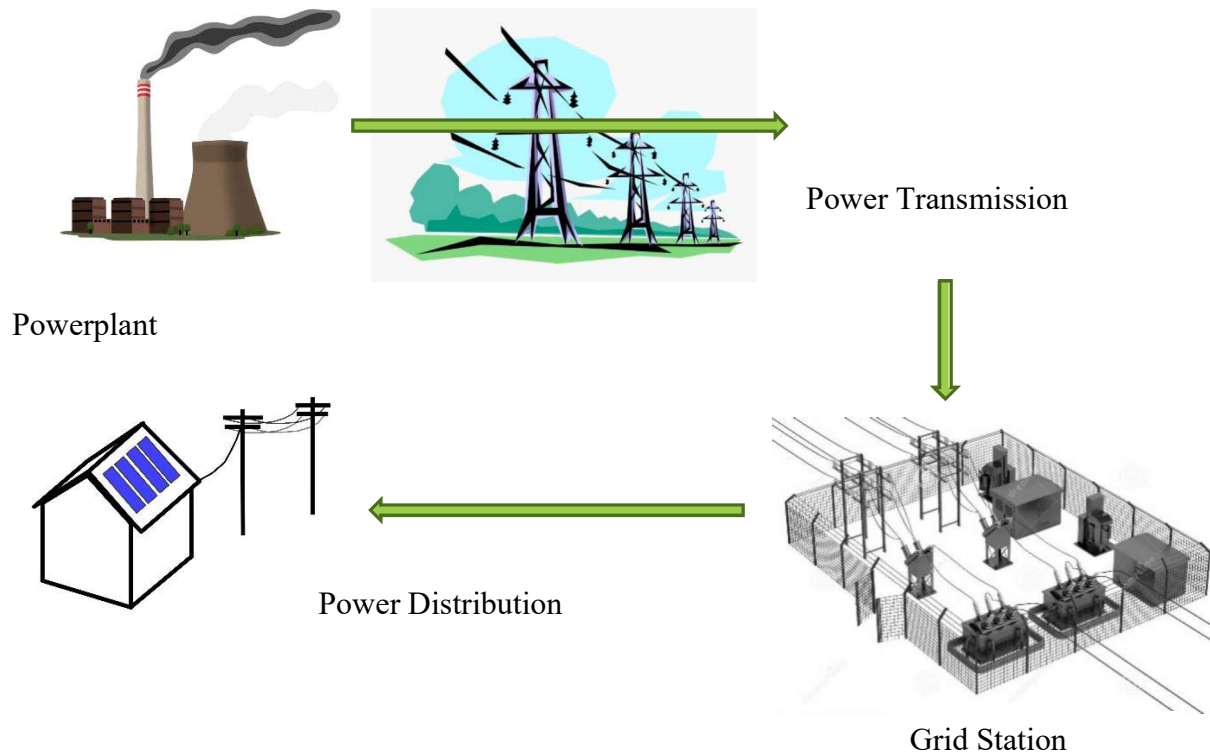


Fig. 1.1 Electricity from generation site to the home

Voltage

A voltage is a potential difference between two points of an electric circuit. A voltage is the pressure from an electrical power source that causes current in an electric circuit. A voltage is like a diffusion phenomenon as shown in Figure 1.2. A voltage is represented by V and is measured in volts (V).

Current

Flow of charges inside an electric circuit is called electric current. An electric power source such as a battery consists of high number of charged particles which creates a potential difference between two points of an electric circuit, causing current. Figure 1.3 shows an analogy between water and current. A current is represented by I and is measured in amperes (A).

Chapter 1

Do you know?

- A voltage is just like diffusion phenomenon in an electric circuit.

Fig. 1.2 Diffusion of Electrons

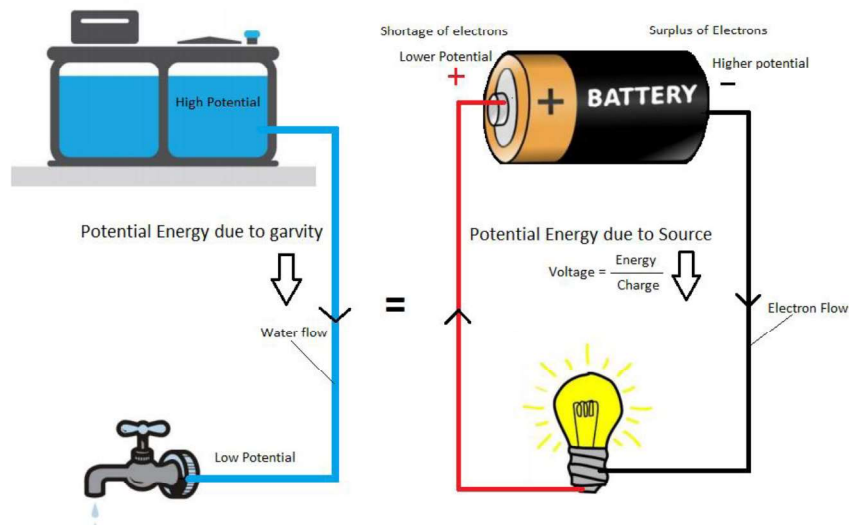


Fig. 1.3 Analogy between Electric Supply and Water Supply

Resistance

A resistance is a component opposing the current flow in an electric circuit. The higher the value of resistance, more will be the opposition. A resistance is represented by R and is measured in ohms (Ω).



Figure 1.4 Resistance

AC and DC Quantities

There are two types of current. Direct Current (DC) and Alternating Current (AC). Magnitude (value) and direction of DC remains constant while AC keeps on changing its direction. Similarly, there exists AC and DC voltages.

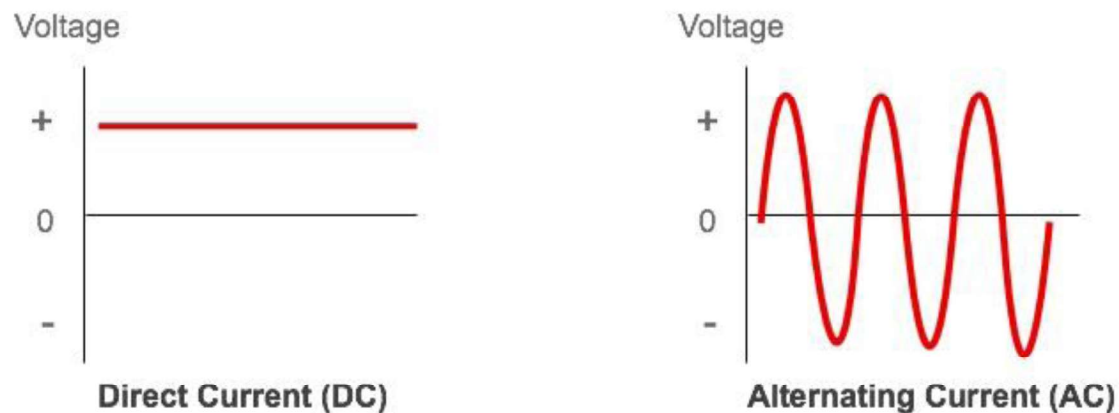


Figure 1.5 Difference between AC and DC

Point to Ponder

- Why there exists two different types of quantities: AC and DC voltage and current?

1.1.3 Conductors, Insulators and Semiconductors

Conductors

Current can flow in conductors. Conductors have free electrons on its surface which allow current to pass through easily. They are made up of atoms having free electrons. Copper, silver and aluminum are good conductors of electricity.

Insulators

Insulators are made up of material that do not allow electricity to pass through it. They are made up of atoms that do not have free electrons. Plastic, glass and rubber etc. are good insulators for electricity.

Chapter 1

Semi conductor

Semiconductors have limited free electrons so their conductivity lies between conductors and Insulators. Silicon and germanium are semiconductors.

Activity

- Identify conductors and insulators of heat in your home.

Point to Ponder

- Is a good conductor of heat also a good conductor of current?

1.1.4 Electrical Power and Energy

Electric Power

Electric power of an electric circuit or appliance is the rate at which it consumes energy. Unit of Power is Watt which is 1 Joule per second. Equation 1 represents power in terms of energy.

$$P = \frac{\Delta E}{\Delta t} \text{ -----(1)}$$

ΔE is work being done. So, equation 1.1 can be rewritten as:

$$P = \frac{W}{Q} \frac{Q}{t} \text{ -----(2)}$$

Voltage is basically the work done (W) per unit charge (Q). While current is charge per unit of time. So, equation 1.2 can be rewritten as:

$$P = VI \text{ -----(3)}$$

Electric Energy

It is a form of energy resulting from flow of electrons. It is usually derived from power. Most common unit of consuming energy is in terms of kWh, that is equivalent of using 1

kilowatt power for one hour. kWh is also used as a unit for computing electricity bill by energy suppliers.

Do you know?

- When purchasing an electric appliance, consumer asks for the power rating (Watts) of the appliance, which is the rate at which it will consume energy.

Activity

- Compute the static load (total power rating of all the appliances in your home).
- From your electric energy meter, note down the units which you have consumed.

1.1.5 Instruments to Measure Electrical Quantities

Ohm Meter

An ohm meter is a device to measure resistance. Connecting leads of ohm meter are connected with the resistance to measure and display the resistance on ohm meter.

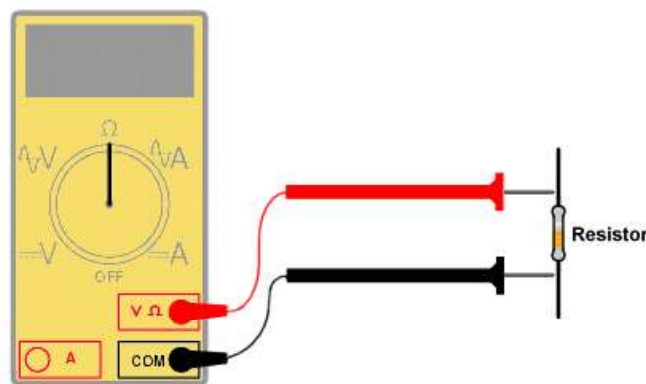


Figure 1.6 Ohm meter

Chapter 1

Voltmeter

A voltmeter is used to measure voltages. Probes of voltmeter are connected with the terminals of the circuits across which voltages are to be measured. A voltmeter is always connected in parallel across a component whose voltages are to be measured.

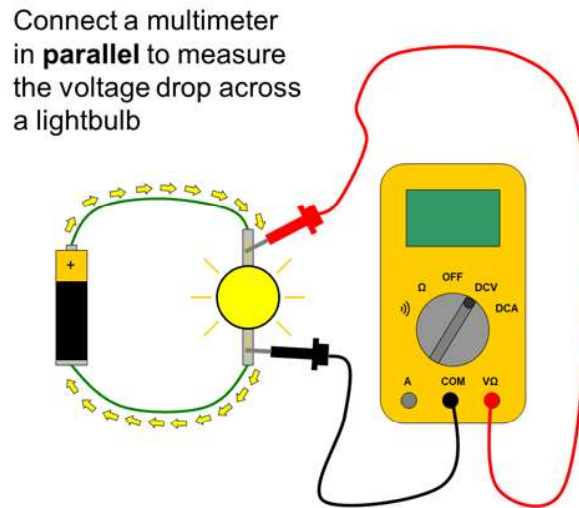


Fig 1.7 Volt meter

Ammeter

An ammeter is used to measure current. Probes of ammeter are connected with the terminals of the circuits across which current is to be measured. It should be noted that ammeter is always connected in series (as shown in Figure 1.8) in the circuit whose current is to be measured.

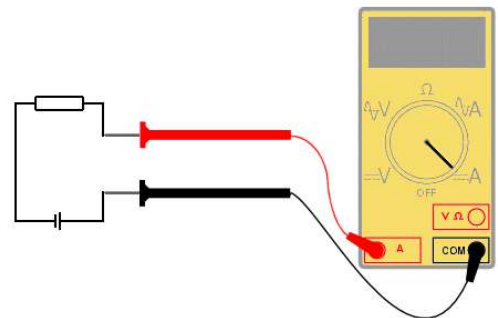


Fig 1.8 Ammeter

Digital Multi meter

A digital multi meter (DMM) is an instrument to measure different electrical quantities. Digital multi meters combine the testing capabilities of single-task meters—the voltmeter (for measuring volts), ammeter (amps) and ohmmeter (ohms).



Fig 1.9 DMM

Oscilloscope

An oscilloscope is a laboratory instrument commonly used to display and analyze the waveform of electronic signals. It draws a graph of the instantaneous signal voltage in terms of time.

There are multiple controls on an oscilloscope. Scale of x-axis and y-axis can be adjusted. There is a separate control to adjust the time scale of x-axis. To scale the y-axis, there is a knob called voltage/div. There are multiple channels on an oscilloscope. This means you can plot different signals simultaneously. To calibrate the oscilloscope, a probe is connected on an input of a channel and the other end is connected to the calibration terminal. The resulting waveform is then calibrated on the required scale.



Fig 1.10 Oscilloscope

1.2 Ohm's Law

Ohm's law states that the current flowing in a circuit is directly proportional to the Voltage across it, given that all the physical quantities and temperature remains the same.

$$V \propto I \text{ -----(4)}$$

Chapter 1

This means that higher the voltage across a component, more will be the current through it. Introducing a constant of proportionality in equation (4):

$$V = IR \text{ -----(5)}$$

Resistance (R) is the constant of proportionality. From equation (5), we can find the third parameter if we know two of them.

Example 1.1

If the resistance of an electric hearth is 50Ω and a current of 3.2 A flows through the resistance. Find the voltage across resistance.

Solution:

If we are asked to calculate the value of voltage with the value of current and resistance given to us, then cover V in the triangle. Now, we are left with I and R or more precisely $I \times R$.

Therefore, we use the following formula to calculate the value of V:

$$V = I \times R$$

Substituting the values in the equation, we get

$$V = 3.2 \text{ A} \times 50 \text{ ohms} = 160 \text{ V}$$

$$\mathbf{V = 160V}$$

Example 1.2

A voltage source of 8.0 V is connected to a purely resistive electrical appliance (a light bulb). An electric current of 2.0 A flows through it. Assume the conducting wires to be resistance-free. Calculate the resistance offered by the electrical appliance.

Solution:

When we are asked to find out the value of resistance when the values of voltage and current are given, then we cover R in the triangle. This leaves us with only V and I, more precisely $V \div I$.

Substituting the values in the equation, we get

$$R = V \div I$$

$$R = 8 \text{ V} \div 2 \text{ A} = 4 \Omega$$

$$\mathbf{R = 4 \Omega}$$

Calculating Electrical Power Using Ohm's Law

Ohm's Law can be used to rewrite equation 1.3. The power formula written in terms of ohm's law is:

$$P = (IR) I = I^2 R \text{ -----(6)}$$

or

$$P = V \frac{V}{R} = \frac{V^2}{R} \text{ -----(7)}$$

Example 1.3

If the current and voltage of an electric circuit are given as 2.5A and 10V respectively. Calculate the electrical power?

Solution: Given measures are,

$$I = 2.5\text{A and } V = 10\text{V}$$

The formula for electric power is,

$$P = VI$$

$$P = 10 \times 2.5 = 25\text{watts}$$

Example 1.4

Calculate the power of an electrical circuit consisting of resistance 3Ω and a current 4A flowing through this circuit?

Solution:

Given parameters are,

$$I = 4\text{A and } R = 3\Omega$$

Electric power formula is,

$$P = I^2 R$$

$$P = 4^2 \times 3$$

$$P = 16 \times 3 = 48 \text{ watts}$$

Chapter 1

1.3 Cells and Batteries

Cell

A cell is a single unit device that converts chemical energy into electrical energy. Typical rating of a cell is 1.5V while 3.7V is also available. A cell can be rechargeable or non-rechargeable.



Cell

Battery

A battery is a group of cells connected together. Cells can be connected in series or parallel. Polarity of cells must be taken care of while forming a battery.

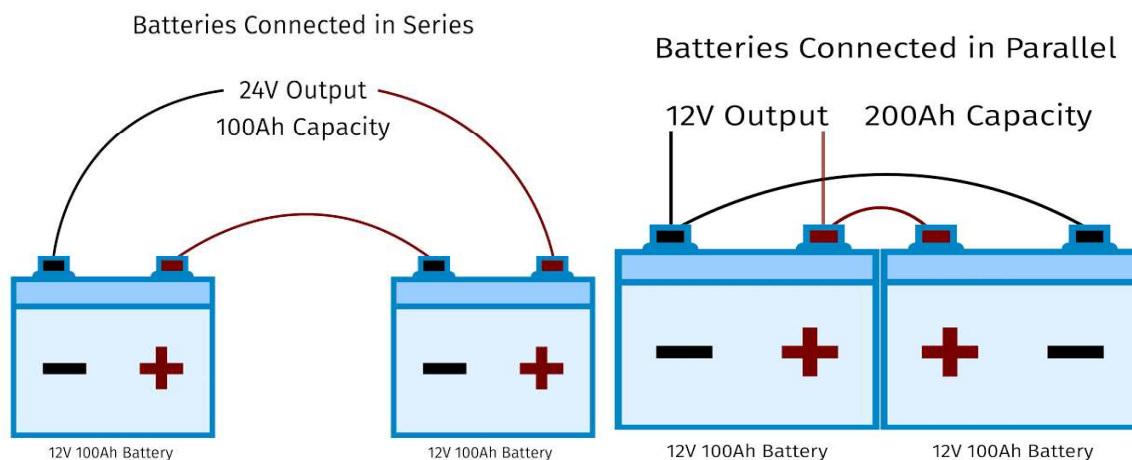


Battery

Series and Parallel Connection of Cells/Batteries

Different batteries can also be tied together to enhance voltage and current capacity.

Series Connection: A series connection is formed when positive terminal of one battery is connected with the negative terminal of the other battery. Series connection increases the voltage capacity of the battery bank but current capacity remains the same.



Series and Parallel Connection of Batteries

Parallel Connection: A parallel connection is formed when positive terminals are connected with positive terminals of other batteries and negative terminals are also

connected together. Parallel connection increases the current capacity of the battery bank while voltage rating remains the same.

Point to Ponder

- What is the voltage rating of a UPS battery?
- Voltage of a battery is DC or AC.

Do you know?

- The voltage rating of a laptop battery is 19.5V and is usually formed of 6 cells.

Importance of Batteries

- ❖ A battery provides portability to a device / appliance.
- ❖ The main purpose of the batteries is to store electrical energy.
- ❖ Rechargeable batteries enable multiple times usage of batteries.
- ❖ Size of battery is very crucial to use it in a specific application.

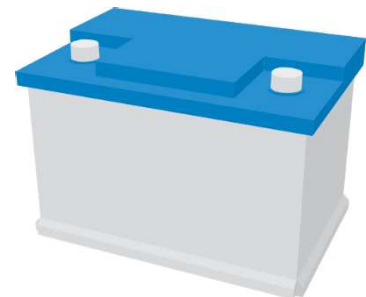
1.3.1 Types of Batteries

There are four main types of secondary batteries:

- ❖ Lead – Acid Batteries
- ❖ Nickel – Cadmium Batteries
- ❖ Nickel – Metal Hydride Batteries
- ❖ Lithium – Ion Batteries

Lead–Acid Batteries: Lead-acid batteries are the most widely used batteries. They have 70 to 80% conversion efficiency. These batteries are mostly used in UPS and vehicles.

Nickel–Cadmium Batteries: Ni-Cd batteries use Nickel Oxyhydroxide (NiOOH) as Cathode and Cadmium metal (Cd) as anode. These batteries usually have a voltage rating of



Lead-Acid Battery



Ni-Cd Battery

Chapter 1

1.2V. These batteries are mostly used in photography equipment and power tools.

Nickel–Metal Hydride Batteries: These are relatively new type of batteries are an extended version of Nickel – Hydrogen Electrode batteries, which were exclusively used in aerospace applications (satellites). The positive electrode is the Nickel Oxyhydroxide (NiOOH) while the negative electrode of the cell is a metal alloy, where hydrogen is stored reversibly.



Ni-MH Battery

Lithium–Ion Batteries: A lithium-ion battery is a family of rechargeable battery types in which lithium ions move from the negative electrode to the positive electrode during discharge and back when charging. They have the best energy to weight ratios.



Lithium-Ion Battery

1.3.2 Components of a Battery

Cells are comprised of 3 essential components.

1. The **Anode** is the negative or reducing electrode that releases electrons to the external circuit and oxidizes during and electrochemical reaction.
2. The **Cathode** is the positive or oxidizing electrode that acquires electrons from the external circuit and is reduced during the electrochemical reaction.
3. The **Electrolyte** is the medium that provides the ion transport mechanism between the cathode and anode of a cell. Electrolytes are often thought of as liquids, such as water or other solvents, with dissolved salts, acids, or alkalis that are required for ionic conduction.

1.3.3 Working Principle of a Battery

A battery works on the oxidation and reduction reaction of an electrolyte with metals. When two dissimilar metallic substances, called electrode, are placed in a diluted electrolyte, oxidation and reduction reaction take place in the electrodes respectively depending upon the electron affinity of the metal of the electrodes. As a result of the

oxidation reaction, one electrode gets negatively charged called *cathode* and due to the reduction reaction, another electrode gets positively charged called *anode*.

The cathode forms the negative terminal whereas anode forms the positive terminal of a battery. To understand the basic principle of battery properly, first, we should have some basic concept of electrolytes and electrons affinity. Actually, when two dissimilar metals are immersed in an electrolyte, there will be a potential difference produced between these metals.

It is found that, when some specific compounds are added to water, they get dissolved and produce negative and positive ions. This type of compound is called an electrolyte. The popular examples of electrolytes are almost all kinds of salts, acids, and bases etc. The energy released during accepting an electron by a neutral atom is known as electron affinity.

If two different kinds of metals are immersed in the same electrolyte solution, one of them will gain electrons and the other will release electrons. The metal with low electron affinity will gain electrons from the negative ions of the electrolyte solution. Whereas, the metal with high electron affinity will release electrons and these electrons come out into the electrolyte solution and are added to the positive ions of the solution. In this way, one of these metals gains electrons and another one loses electrons. Therefore, there will be a difference in electron concentration between these two metals.

This difference in electron concentration causes an electrical potential difference developed between the metals. This electrical potential difference or EMF can be utilized as a source of voltage in any electronics or electrical circuit. This is a general and basic working principle of a battery.

Chapter 1

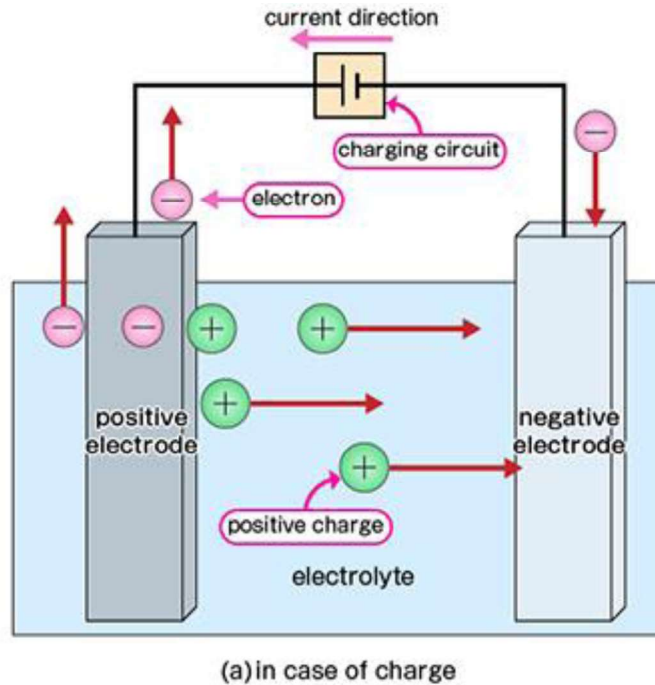


Fig 1.11 Construction and Working of a Battery

1.3.4 Charging Principle of a Battery

During charging of a battery, external DC source is applied to the battery. The negative terminal of the DC source is connected to the negative plate or anode of the battery and positive terminal of the source is connected to the positive plate or cathode of the battery. Now, due to external DC source, electrons will be injected in the anode. Reduction reaction takes place in the anode instead of cathode. Actually, in case of discharge of the battery, reduction reaction takes place at cathode. Due to this reduction reaction, the anode will regain electrons and returns to its previous state when the battery was not discharged.

As the positive terminal of the DC source is connected to the cathode, the electrons of this electrode will be attracted by this positive terminal of DC source. As a result, oxidation reaction takes place at the cathode and cathode material regains its previous state (when it was not discharged). This is the overall basic of charging of battery.

Note for lecture.

➤ Charging and discharging curves of a battery can also be explained here.

1.3.5 Maintaining a Battery

Maintenance increases battery life. Follow the following procedures for better maintaining a battery.

- Maintain electrolyte level of a battery.
- Regularly clean the terminals of a battery
- Battery should be charged slowly
- Allow the battery to cool down once it is fully charged.

1.3.6 Testing of a Battery

Visual Test

- Check the container, cover and terminals. If physical damage is present, replace the battery.
- Check the indicator (If the battery has the indicator). Always look right down when viewing the indicator and lightly tap the indicator on the battery to dislodge any air bubbles.

Voltage Test

- If voltage is below 12.4V, recharge the battery immediately.

Load Test

- Connect the battery tester to battery terminals.
- Measure the temperature of the battery around.
- Apply the load for 15 seconds and read the voltage.
- If the voltages drop significantly, recharge the battery and test again.
- If the battery fails the load test twice, replace it.

Chapter 1

Gravity Test

The most accurate and direct way to test the state of charge of a battery cell is to determine the specific gravity of the battery electrolyte. The higher the specific gravity of the electrolyte the higher the state of charge.

Hydrometer: Hydrometer is used to perform gravity test. Hydrometers come in many sizes and shapes. A hydrometer with a float is recommended, contained in a glass vessel with a rubber bulb to draw the acid into the tube. Stay away from floating-colored balls as the extra inaccuracy results in very subjective testing. The hydrometer should give you a numeric reading directly from the instrument. A good hydrometer is accurate to +/- 0.005 points so 1.265 could read from 1.260-1.270. The instrument accuracy should be known.

Test

- It is recommended to disconnect the battery especially if on a high rate of charge/discharge.
- Remove vent cap. Carefully insert the hydrometer into the cell, not pushing down on the top of the plates.
- Carefully draw liquid into the hydrometer and avoid "bumping" the hydrometer. Be careful the float is not flooded (too much liquid) or sticking to the sides of the glass tube.
- Obtain a reading by looking directly at the float.
- Repeat steps 3-5 to reconfirm reading.
- Record the cell number and result.

Tools for Testing a Battery

If you have a non-sealed battery, it is highly recommended that you use a good quality temperature compensating hydrometer. There are two basic types of hydrometers, the floating ball, and the gauge. The gauge type tends to be much easier to read and doesn't involve the need to decipher colored balls. Battery hydrometers can be purchased at an auto parts or battery store.

To test a sealed battery or to troubleshoot a charging or electrical system, you will need a digital voltmeter with 0.5 percent (or better) accuracy. A digital voltmeter can be purchased at an electronics store. Analog (needle type) voltmeters are not accurate enough to measure the millivolt differences of a battery's state-of-charge or measure the output of the charging system. A battery load tester is optional.

1.3.7 Battery Life

A 10 Ah battery delivering 1A, would last 10 hours. If delivering 10A, it would last for only 1 hour, or if delivering 5A, it would last only for 2 hours.

In other words, you can have “any time” as long as when you multiply it by the current, you get 10Ah (the battery capacity).

So, no more confusion on how to calculate battery life.

For a 18650 2500mAh(2.5Ah) battery with a device that draws 500mA(0.5A) you have:
 $2.5\text{Ah}/0.5\text{A}=5\text{ Hours}$

Key Points

- Electricity is the secondary form of energy which means that other sources of energy i.e., kinetic energy, thermal energy and solar energy etc. are converted into electrical energy.
- The electricity that power plants generate is delivered to consumers over transmission and distribution power lines.
- A voltage is a potential difference between two points of an electric circuit.
- Flow of charges inside an electric circuit is called electric current.
- Magnitude and Direction of DC remains constant while AC keeps on changing its direction.
- Conductors have free electrons on its surface which allow current to pass through easily.
- Semiconductors have limited free electrons so their conductivity lies between conductors and Insulators.
- Electric power of an electric circuit or appliance is the rate at which it consumes energy.
- $P = VI$
- Most common unit of consuming energy is in terms of kWh, that is equivalent of using 1 kilowatt power for one hour.
- An ohm meter is a device to measure resistance.
- A voltmeter meter is a device to measure voltage.
- An ammeter is used to measure current.
- $V = IR$
- $P = (IR) I = I^2 R$
- $P = V \frac{V}{R} = \frac{V^2}{R}$
- There are three main components of a battery: Anode, Cathode and Electrolyte.

Exercise

Multiple Choice Questions

1. An ohm meter is used to measure
 - a. Resistance
 - b. Voltage
 - c. Current
 - d. None of these
2. A voltmeter is used to measure
 - a. Resistance
 - b. Voltage
 - c. Current
 - d. None of these
3. An ammeter is used to measure
 - a. Resistance
 - b. Voltage
 - c. Current
 - d. None of these
4. In series connection following remains the same
 - a. Resistance
 - b. Voltage
 - c. Current
 - d. None of these
5. In parallel connection following remains the same
 - a. Resistance
 - b. Voltage
 - c. Current
 - d. None of these

Write short answer of the following

1. Define Electricity.
2. How is electricity transmitted from power plants to your homes?
3. Define Voltage, Resistance and Current.
4. What is the difference between AC and DC quantities?
5. Differentiate between conductors, insulators and semiconductors.

Answer the following question in detail

1. Differentiate between electric power and electric energy.
2. Describe Ohm meter.
3. Describe Ammeter.
4. Describe Voltmeter.
5. Differentiate between cells and batteries.

Solve the Following Problems

1. If the resistance of an electric hair dryer is $80\ \Omega$ and a current of $2.2\ \text{A}$ flows through the resistance. Find the voltage between two points.
2. An EMF source of $10.0\ \text{V}$ is connected to a purely resistive electrical appliance. An electric current of $1.5\ \text{A}$ flows through it. Consider the conducting wires to be resistance-free. Calculate the resistance offered by the electrical appliance.
3. If the current and voltage of an electric circuit are given as $3.5\ \text{A}$ and $9\ \text{V}$ respectively. Calculate the electrical power?
4. Calculate the power of an electrical circuit consisting of resistance $6\ \Omega$ and a current $4.5\ \text{A}$ flowing through this circuit?
5. Calculate the power of an electrical circuit consisting of resistance $7\ \Omega$ and a voltage of $20\ \text{V}$ across it?